

TUTORIUM-7

T7.2: Geg: $f: \mathbb{R} \rightarrow \mathbb{R}$ stetig - (I)

$$\lim_{x \rightarrow -\infty} f(x) = -\infty \quad - (II)$$

$$\lim_{x \rightarrow \infty} f(x) = \infty \quad - (III)$$

z.z.: f surjektiv, d.h.

$$\forall y \in \mathbb{R} \exists x \in \mathbb{R}: y = f(x)$$

Bew.: III $\Leftrightarrow \forall R > 0 \exists K > 0: \forall x > R: f(x) > K$
 II $\Leftrightarrow \forall \tilde{R} < 0 \exists \tilde{K} < 0: \forall x < \tilde{R}: f(x) < \tilde{K}$

Sei $\xi \in \mathbb{R}$ bel.

z.z. ist: $\exists x_0: f(x_0) = \xi$

$$\left. \begin{array}{l} \rightarrow \exists R_+ > 0: f(R_+) \geq \xi \\ \rightarrow \exists R_- < 0: f(R_-) \leq \xi \end{array} \right\} \Leftrightarrow \xi \in [f(R_-), f(R_+)]$$

Zwischenwertsatz: Ist $f: [a, b] \rightarrow \mathbb{R}$ stetig, dann

$$\forall y \in [f(a), f(b)] \exists x_0 \in [a, b]: f(x_0) = y$$

Aus zws. folgt: $\forall \xi$

$$\exists x_0 \in [R_-, R_+]: f(x_0) = \xi$$

T7.1: Landau Symbole:

(a) $\overline{D} = D \Leftrightarrow D$ Abg. ; $D^\circ = D \Leftrightarrow$ offene.

$f, g: D \xrightarrow{\mathbb{R}} \mathbb{R}$, mit $x_0 \in \overline{D}$

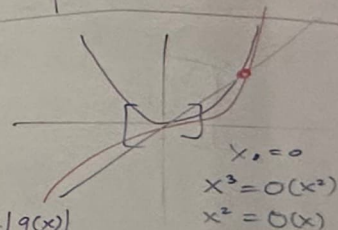
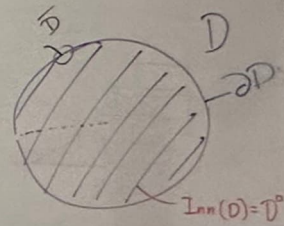
$$f(x) = O(g(x)) \quad (x \rightarrow x_0)$$

$$\exists \delta > 0 \exists c \in \mathbb{R} \forall x \in D: |x - x_0| < \delta \Rightarrow |f(x)| \leq c \cdot |g(x)|$$

Aufgabe:

$$\lim_{x \rightarrow x_0} \left| \frac{f(x)}{g(x)} \right| =: b \in (0, \infty)$$

$$\text{z.z. } f(x) = O(g(x)) \wedge g(x) = O(f(x))$$



Wähle

$$\varepsilon = \frac{b}{2} \quad \exists \delta > 0 : \forall x \in D : |x - x_0| < \delta \Rightarrow \left| \frac{f(x)}{g(x)} - b \right| < \varepsilon$$

$$- \varepsilon + b < \left| \frac{f(x)}{g(x)} \right| < b + \varepsilon$$

$$\rightarrow c = b + \varepsilon = \frac{3b}{2} \Rightarrow |f(x)| < c \cdot |g(x)| \Leftrightarrow f(x) = O(g(x))$$

$$\rightarrow \frac{1}{c} - \varepsilon + b = \frac{b}{2} \Rightarrow |g(x)| < \tilde{c} \cdot |f(x)| \Leftrightarrow g(x) = O(f(x))$$

$$(b) \lim_{x \rightarrow 0} \frac{\sin x}{x} = \lim_{x \rightarrow 0} \frac{x - \frac{x^3}{6} + \frac{x^5}{120} - \dots}{x} = \lim_{x \rightarrow 0} 1 - \frac{x^2}{6} + \dots = 1$$

$$\sin(x) = O(x) \quad \wedge \quad x = O(\sin(x)) \quad (x \rightarrow 0)$$

$$|\sin(x)| \leq 1 \Rightarrow \sin(x) = O(1) \quad (x \rightarrow \infty)$$

$\forall x$

$$(c) \sinh(x) = \frac{e^x - e^{-x}}{2}$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$\lim_{x \rightarrow 0} \left(\frac{\sinh(x)}{x} \right) = \lim_{x \rightarrow 0} \left(\frac{e^x - e^{-x}}{2x} \right) = \lim_{x \rightarrow 0} \frac{e^x - 1}{2x} + \lim_{x \rightarrow 0} \frac{e^{-x} - 1}{-2x}$$

$$= \lim_{x \rightarrow 0} \left(\frac{1 + \cancel{x} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots - 1}{2x} \right) + \lim_{x \rightarrow 0} \left(\frac{1 - \cancel{x} + \frac{x^2}{2!} + \frac{-x^3}{3!} + \dots - 1}{-2x} \right)$$

$$= \lim_{x \rightarrow 0} \frac{1}{2} + \frac{x}{2 \cdot 2} + \frac{x^2}{2 \cdot 3!} + \dots + \lim_{x \rightarrow 0} \frac{1}{2} = \frac{1}{2} + \frac{1}{2} = 1$$

$$\sinh(x) = O(x) \quad (x \rightarrow 0)$$

$$\lim_{x \rightarrow \infty} \left(\frac{\sinh(x)}{e^x} \right) = \lim_{x \rightarrow \infty} \left(\frac{e^x - e^{-x}}{2e^x} \right) = \lim_{x \rightarrow \infty} \left(\frac{1}{2} - \frac{e^{-2x}}{2} \right) = \frac{1}{2} - \lim_{x \rightarrow \infty} \frac{e^{-2x}}{2} = \frac{1}{2}$$

$$\sinh(x) = O(e^x) \quad (x \rightarrow \infty)$$

Ableitungen: Diffbarkeit im Punkte / auf nem Intervall.

$f: \mathbb{R} \supseteq I \rightarrow \mathbb{R}$ ist im Punkte $x_0 \in I$ diffbar, gdw.

Ableitung
 $f'(x_0) := \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}$ existiert. (*)

$\left(\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \right)$ „

→ Diff auf dem Intervall I : Wenn $\forall x_0 \in I$: $(*)$ ($f'(x_0)$ existiert)
 f Diffbar. (auf I).

(a) $\sin: \mathbb{R} \rightarrow \mathbb{R}, x \mapsto \sin(x)$; $\frac{\sin(x) - \sin(x_0)}{x - x_0} = \frac{\sin(x)}{x}$

- Für $x_0 = 0$: $\lim_{x \rightarrow 0} \frac{\sin(x) - \sin(0)}{x - 0} = \lim_{x \rightarrow 0} \frac{\sin(x)}{x} \stackrel{T.1.1(b)}{=} 1$

- Für $x_0 \in \mathbb{R} \setminus \{0\}$: $\lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin(x)}{h} = \lim_{h \rightarrow 0} \frac{\sin x \cosh + \cos x \sinh - \sin x}{h}$

(c) $\sinh(x) = \frac{e^x - e^{-x}}{2}$

$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$

$\lim_{x \rightarrow 0} \left(\frac{\sinh(x)}{x} \right) = \lim_{x \rightarrow 0} \left(\frac{e^x - e^{-x}}{2x} \right) = \lim_{x \rightarrow 0} \frac{e^x - 1}{2x} + \lim_{x \rightarrow 0} \frac{e^{-x} - 1}{-2x}$

$= \lim_{x \rightarrow 0} \left(\frac{1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots - 1}{2x} \right) + \lim_{x \rightarrow 0} \left(\frac{1 - x + \frac{x^2}{2!} + \frac{-x^3}{3!} + \dots - 1}{-2x} \right)$

$= \lim_{x \rightarrow 0} \frac{1}{2} + \frac{x}{2! \cdot 2} + \frac{x^2}{2 \cdot 3!} + \dots + \lim_{x \rightarrow 0} \frac{1}{2} = \frac{1}{2} + \frac{1}{2} = 1$

$\sinh(x) = \theta(x) \quad (x \rightarrow 0)$

$\lim_{x \rightarrow \infty} \left(\frac{\sinh(x)}{e^x} \right) = \lim_{x \rightarrow \infty} \left(\frac{e^x - e^{-x}}{2e^x} \right) = \lim_{x \rightarrow \infty} \left(\frac{1}{2} - \frac{e^{-2x}}{2} \right) = \frac{1}{2} - \lim_{x \rightarrow \infty} \frac{e^{-2x}}{2} = \frac{1}{2}$

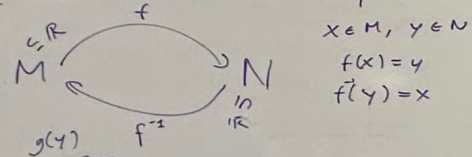
$\sinh(x) = \theta(e^x) \quad (x \rightarrow \infty)$

$\cos x \lim_{h \rightarrow 0} \frac{\sinh h}{h} + \sin(x) \lim_{h \rightarrow 0} \frac{\cosh h - 1}{h}$

$= \cos x + \sin x \cdot \lim_{h \rightarrow 0} \left(1 - \frac{h^2}{2!} + \frac{h^4}{4!} - \dots - 1 \right) = \cos(x)$

$\frac{e^{ix} + e^{-ix}}{2i}$

Recap: Abl. von der Umkehrfkt.



$$\underbrace{f(f^{-1}(y))}_{=y} = y \Leftrightarrow \frac{d}{dy} f(g(y)) = 1$$

$$\frac{d}{dg(y)} f(g(y)) \cdot \frac{d}{dy} g(y) = 1 \Rightarrow f'(g(y)) \cdot g'(y) = 1$$

$$\Rightarrow \boxed{[f^{-1}(y)]' = \frac{1}{f'(f^{-1}(y))}} \quad \text{formelzettel}$$

$$(b) \arcsin(x) = \frac{1}{\sin'(\arcsin(x))} = \frac{1}{\cos(\arcsin(x))}$$

$$\cos(\beta) = \sqrt{1 - \sin^2 \beta} \quad \left| \begin{array}{l} \cos^2 \beta + \sin^2 \beta = 1 \\ \end{array} \right.$$

$$= \frac{1}{\sqrt{1 - (\sin(\arcsin(x)))^2}}$$

$$= \frac{1}{\sqrt{1 - x^2}}$$

Tutorium-7

$$\begin{aligned} (c) \frac{d}{dx} x^\alpha &= \frac{d}{dx} e^{\alpha \cdot \ln(x)} = e^{\alpha \cdot \ln(x)} \cdot [\alpha \ln(x)]' \\ &= e^{\alpha \ln(x)} \cdot \alpha \cdot \frac{1}{x} = x^\alpha \cdot \alpha \cdot \frac{1}{x} = \alpha \cdot x^{\alpha-1} \end{aligned}$$

$$d) \sqrt{1-x^2} = f \circ g(x) \quad , \quad f(y) = \sqrt{y} \quad ; \quad g(x) = 1-x^2$$

$$f'(y) = \frac{1}{2\sqrt{y}} \quad ; \quad g'(x) = -2x \quad , \quad y = g(x)$$

$$(f \circ g(x))' = f'(g(x)) g'(x) = \frac{-2x}{2\sqrt{1-x^2}} = \frac{-x}{\sqrt{1-x^2}}$$

$$\frac{d}{dx} f(x) = f'(x)$$